



Introduction to data integrity

Hi! Good to see you! My name is Sally, and I'm here to teach you all about processing data. I'm a measurement and analytical lead at Google. My job is to help advertising agencies and companies measure success and analyze their data, so I get to meet with lots of different people to show them how data analysis helps with their advertising.

Speaking of analysis, you did great earlier learning how to gather and organize data for analysis. It's definitely an important step in the data analysis process, so well done!

Now let's talk about how to make sure that your organized data is complete and accurate. Clean data is the key to making sure your data has integrity before you analyze it. We'll show you how to make sure your data is clean and tidy. Cleaning and processing data is one part of the overall data analysis process. As a quick reminder, that process is Ask, Prepare, Process, Analyze, Share, and Act. Which means it's time for us to explore the Process phase, and I'm here to guide you the whole way. I'm very familiar with where you are right now.

I'd never heard of data analytics until I went through a program similar to this one. Once I started making progress, I realized how much I enjoyed data analytics and the doors it could open. And now I'm excited to help you open those same doors! One thing I realized as I worked for different companies, is that clean data is important in every

industry. For example, I learned early in my career to be on the lookout for duplicate data, a common problem that analysts come across when cleaning. I used to work for a company that had different types of subscriptions. In our data set, each user would have a new row for each subscription type they bought, which meant users would show up more than once in my data.

So if I had counted the number of users in a table without accounting for duplicates like this, I would have counted some users twice instead of once. As a result, my analysis would have been wrong, which would have led to problems in my reports and for the stakeholders relying on my analysis. Imagine if I told the CEO that we had twice as many customers as we actually did!? That's why clean data is so important. So the first step in processing data is learning about data integrity. You will find out what data integrity is and why it is important to maintain it throughout the data analysis process. Sometimes you might not even have the data that you need, so you'll have to create it yourself.

This will help you learn how sample size and random sampling can save you time and effort. Testing data is another important step to take when processing data. We'll share some guidance on how to test data before your analysis officially begins. Just like you'd clean your clothes and your dishes in everyday life, analysts clean their data all the time, too. The importance of clean data will definitely be a focus here. You'll learn data cleaning techniques for all scenarios, along with some pitfalls to watch out for as you clean. You'll explore data cleaning in both spreadsheets and databases, building on what you've already learned about spreadsheets.

We'll talk more about SQL and how you can use it to clean data and do other useful things, too. When analysts clean their data, they do a lot more than a spot check to make sure it was done correctly. You'll learn ways to verify and report your cleaning results. This includes documenting your cleaning process, which has lots of benefits that we'll explore. It's important to remember that processing data is just one of the tasks you'll complete as a data analyst. Actually, your skills with cleaning data might just end up being something you highlight on your resume when you start job hunting. Speaking of resumes, you'll be able to start thinking about how to build your own from the perspective of a data analyst.

Once you're done here, you'll have a strong appreciation for clean data and how important it is in the data analysis process. So let's get started!